Network Management Laboratory Exercise 5.0: Firewall and DNS configurations.

Objectives:

Configure the DNS and Firewall Server.
Implement protocol firewall level.
Verify connections from other network and check if the firewall is working.
Understand how a DNS server work.
Troubleshoot DNS issues.

Part 1: Firewall configuration

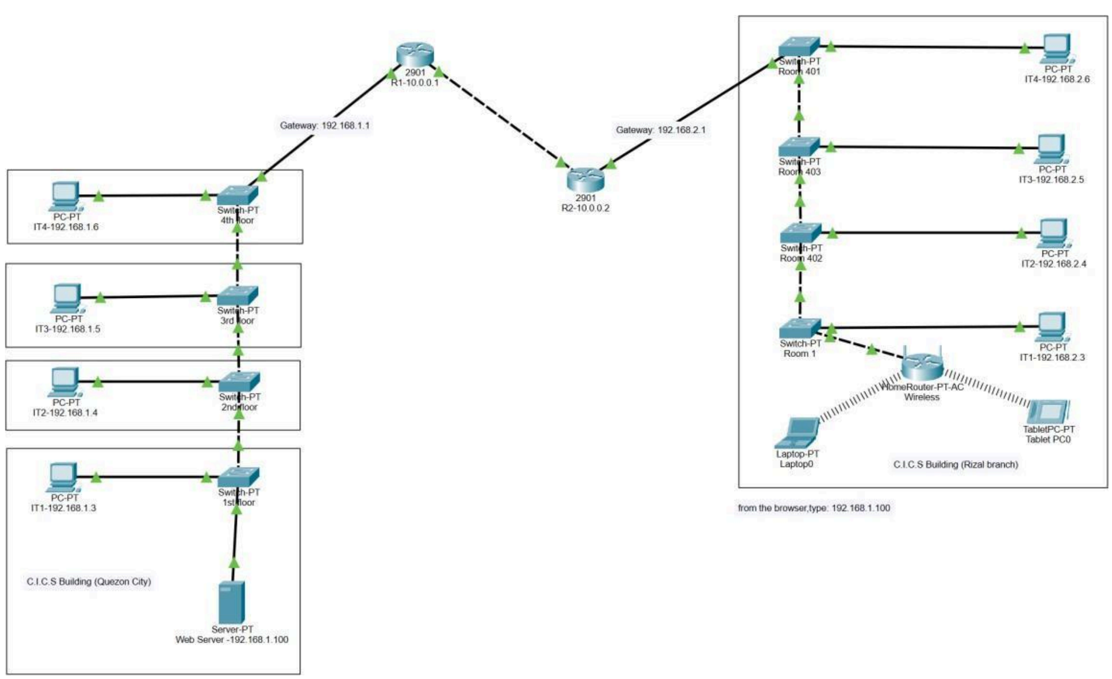
A firewall is a hardware or software network security device that monitors all incoming and outgoing traffic based on a defined set of security rules, it accepts, rejects, or drops that specific traffic.

Accept: Allow traffic.
Reject: Block traffic but respond with “reachable error”.

Drop: Block unanswered traffic firewall establishes a barrier between secure internal networks and untrusted external networks, such as the Internet.

Topology:

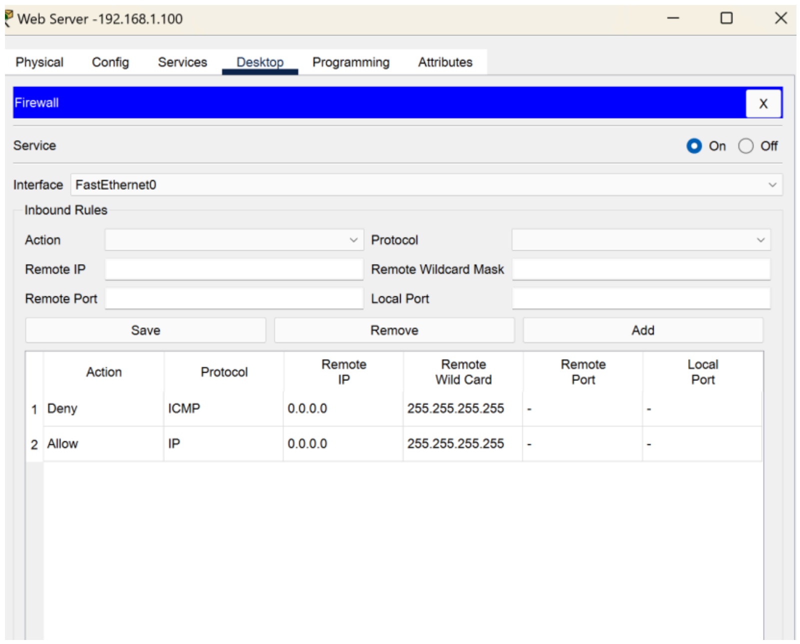
Image of network topology with routers, switches, computers, laptop, tablet, and web server.



Configuring the firewall in a server and blocking packets and allowing web browser.

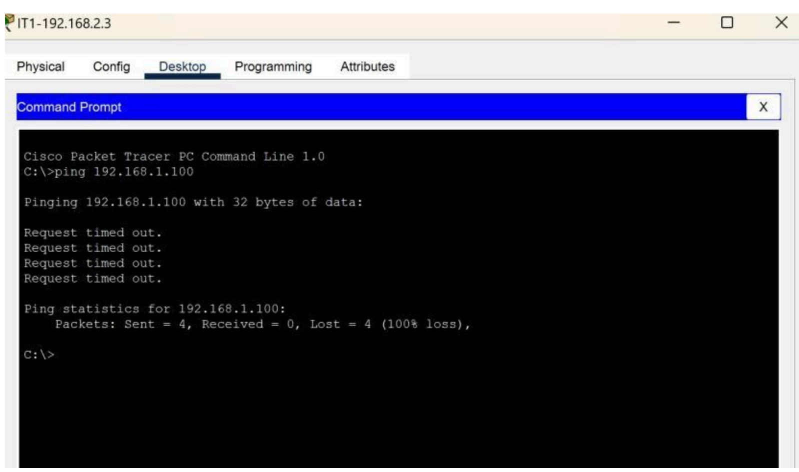
- Click on server-PT then go to the desktop.
- Then click on firewall IPv4.

- Turn on the services.
- First, Deny the ICMP protocol and set remote IP to 0.0.0.0 and Remote wildcard mask to 255.255.255.255.
- Then, allow the IP protocol and set remote IP to 0.0.0.0 and Remote wildcard mask to 255.255.255.255.
- And add them.

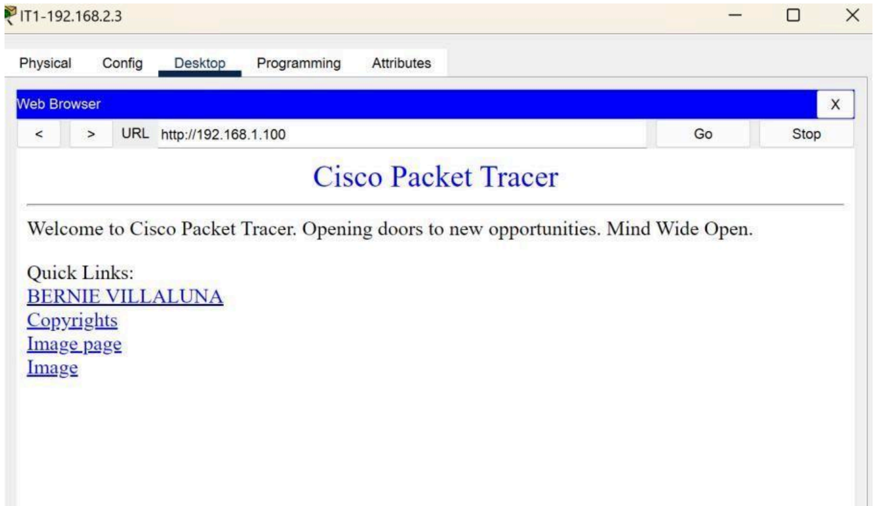


Verifying the network by pinging the IP address of any PC.

- We will use the ping command to do so.
- First, Select any PC from the other branch then go to the command prompt program.
- Then type ping <IP address of targeted node>.
- We will ping the IP address of the Webserver.
- As we can see in the below image we are getting no replies which means the packets are blocked.



Check the web browser by entering the IP address in the URL.

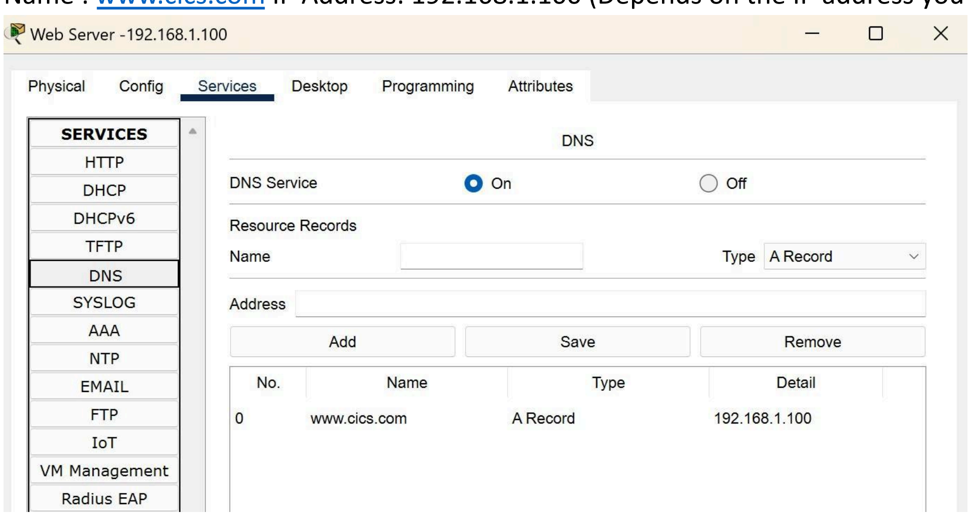


Part 2: DNS Server Configuration.

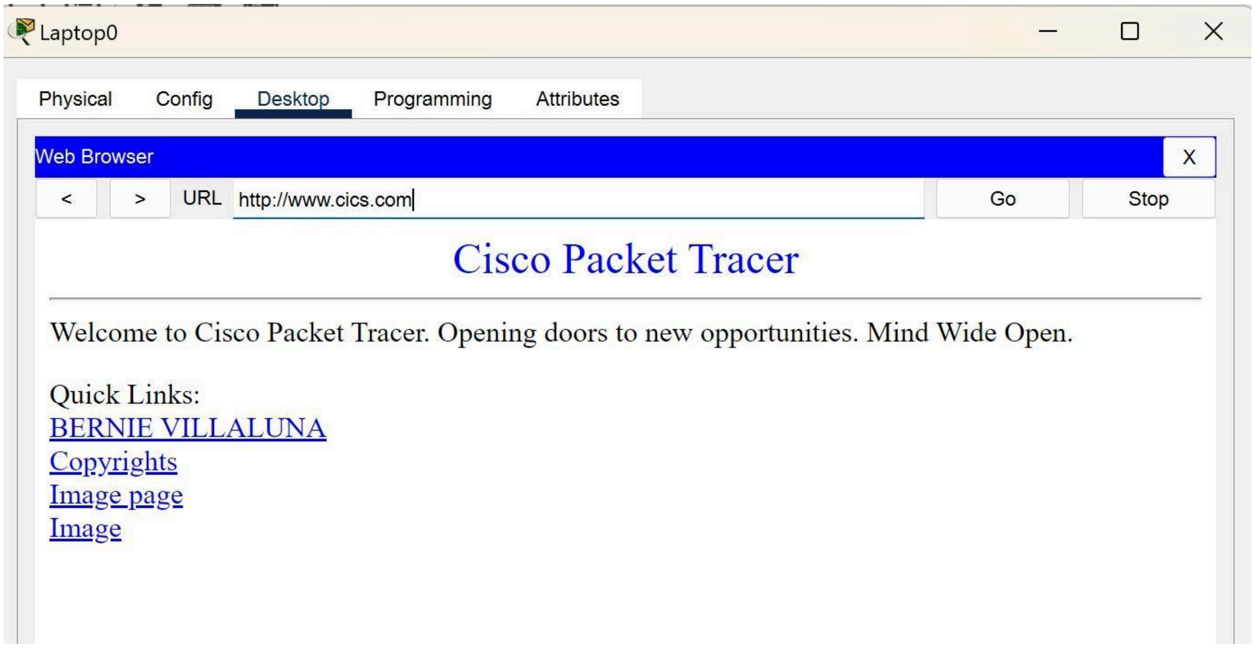
A Domain Name System (DNS) server resolves host names into IP addresses. Although we can access a network host using its IP address, DNS makes it easier by allowing us use domain names which are easier to remember. For example its much easier to access google website by typing http://www.google.com as compared to typing http://208.117.229.214. In either case, you'll access google website, but using domain name is obviously easier.

Configuring the DNS in a server.

- Click on the Server, then Click on Services tab. Click on DNS server from the menu. First turn ON the DNS service, then define names of the hosts and their corresponding IP addresses.
- Name : www.cics.com IP Address: 192.168.1.100 (Depends on the IP address you configured)



- Test domain name – IP resolution. Open the web browser from the laptop computer and type the domain name from the URL address bar. If the DNS service is turned on and all IP configurations are okay, then web browser should work. Don't forget to setup the DNS from the Wi-FI Router.



Reporting:

1. Document the configuration changes made throughout the lab.
2. Screenshot the testing you made using the Ping and Web browser tool.
3. Analyze the results of your tests and troubleshooting efforts.
4. Discuss the learnings and key takeaways from the lab exercise.